

Aiden Khanna
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Education

Duke University *Incoming Class of 2030*
BSE in Electrical and Computer Engineering

New Trier High School *Class of 2026*
GPA: 3.91/4 **ACT:** 36

Stanford Pre-Collegiate Summer Institutes — Ethics in Technology *June 2023*
• Analyzed AI bias, Human Computer Interaction, and Value Sensitive Design; presented on societal impacts of technical systems

Projects

Custom CPU Design — Northwestern University Center for Talent Development *July 2023 - August 2023*
• Led a small team in designing a custom 8-bit CPU with original PCB layouts; assembled and validated hardware prototypes
• Implemented an asynchronously decrementing register for reverse memory sequencing
• Wrote and optimized instruction set architectures using custom assembly opcodes and microcode
• Achieved lowest cycle count (56 cycles) for computing reversed 8-term Fibonacci sequence among peer designs via optimized control logic and asynchronous register design

IoT Musician Assistant — Northwestern University Center for Talent Development *June 2022 - July 2022*
• Collaborated with a small team to engineer an end-to-end IoT system on the Particle Argon for audio playback, auditory and visual metronome, and practice session timing
• Integrated cloud services to automatically generate and manage practice schedules via Google Calendar API
• Implemented real-time device control and monitoring using I2C, MQTT, BLE, and HTTPS
• Designed hardware–software interface coordinating embedded firmware, external hardware, and mobile UI (Blynk)

Experience

Cybersecurity Intern — Cyera *June 2025 - August 2025*
• Designed and implemented an external attack surface monitoring system covering 12,000+ subdomains using a scalable Python pipeline; integrated DNS enumeration, IP resolution, and port scanning, with parallel processing to improve throughput and reduce execution time
• Conducted vendor AI risk assessments using Panorays, evaluating data handling and model usage practices
• Simulated phishing attack scenarios using Wizer to assess internal security posture and response readiness

Intelligent Automation Intern — ITsavvy *June 2024 - August 2024*
• Automated technology spend analysis using Terzo workflows; projected \$320K in annual savings and \$32K/year in setup costs avoided
• Migrated and secured SharePoint knowledge base for 500,000+ student Chromebooks; projected 25% faster training and 15% fewer quality control errors
• Integrated calendar data via Power Automate to automate training metrics collection, improving visibility and scalability

Leadership and Activities

President and Co-Founder — ACK Foundation | ackfoundation.org *October 2025 - Present*
• Raised over \$80,000 for cancer research, environmentalism, and art outreach by organizing a community mural painting in collaboration with youth development program Farmers Rising and through online fundraising

Founder and Chief Editor — Beyond the Sound | beyondthesound.net *October 2023 - Present*
• Founded a multimedia publication focused on music critique and experimental commentary; designed the website and visual assets, published 30+ articles, and conducted interviews with musicians and industry professionals

Other Activities and Interests:

- Guitar (10+ years; recorded and produced a solo EP), music production, Eagle Scout (Boy Scouts of America), running, Latin (biliterate), 3D printing, and cooking